

A novel MCMC-based hydraulic fracture diagnostics approach using water hammer data

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ABSTRACT: The water hammer signatures, which can be observed after pump shutdown at the end of a hydraulic fracture treatment, have proven to contain useful information about the geometry of generated hydraulic fractures. This paper presents a novel hydraulic fracture diagnostics approach using the water hammer data. A water hammer model that describes the pressure transient in hydraulically fractured horizontal well is established. In the model, a new multi-cluster hydraulic fracture boundary condition is derived by considering the perforation friction and leak-off in fractures. The method of characteristics is applied to efficiently solve the water hammer model. The water hammer signals can be effectively reproduced under different fracture characteristics. The MCMC method is applied to construct the posterior statistics of fracture parameters (e.g., width, length, height). The proposed approach is first verified by a series of synthetic models, and then applied to a real field case of a multi-stage hydraulic-fractured horizontal well in a shale gas reservoir to demonstrate its applicability and efficiency.

1. INTRODUCTION

Hydraulic fracturing has been the commonly used stimulation technique for the economic development of shale/tight gas and oil reservoirs. Although rapid progress has been made in fracturing technology, accurate characterization of hydraulic fracture is a great challenge which hinders the evaluation of fracturing efficiency and prediction of well productivity.

Various techniques have been developed to characterize hydraulic fractures, which can be divided into two categories, i.e., direct monitoring methods and indirect interpretation methods. Direct monitoring methods mainly include micro-seismic monitoring (Le Calvez et al. 2007; Zhu et al. 2017), fiber-optic sensing (Liu et al. 2020; Liu et al. 2023), coring observation (Maity et al. 2018), and downhole imaging (Roberts et al. 2018a; Roberts, et al. 2018b). These methods are usually characterized by their high accuracy, but limited by the high cost and limited observation range. Instead, indirect interpretation methods, including flowback analysis (Chen et al. 2018; Liu et al. 2019), pressure transient analysis (He et al. 2017; Wang et al. 2021), and water hammer analysis (Carey et al. 2015; Hu et al. 2022), are commonly used low-cost fracture diagnosis techniques. However, flowback analysis and pressure transient analysis usually require the production data after the well completion, which can

hardly provide the real-time fracture information during hydraulic fracturing. The water hammer analysis has the advantages of short testing time, low-cost and real time.

Water hammer refers to the pressure wave generated by the sudden change of fluid velocity when shutting down surface pump at the end of hydraulic fracturing stimulation. This process can be affected by the friction along the wellbore, pressure loss through perforations, and leak-off into the fractured zone. Once the wellbore geometry and the fluid properties are known, the characteristics of water hammer signatures can reflect the properties of generated hydraulic fractures. Water hammer interpretation may be used as an inexpensive and efficient diagnostic technique for the characterization of hydraulic fracture geometry. Efforts have been paid to forward modeling of the water hammer signatures and infer the hydraulic fracture properties. Holzhausen and Gooch (1985) firstly introduced the water hammer signature into fracture diagnostics, and they used hydraulic impedance testing to analyze the water hammer data and characterize the fracture geometry. Mondal (2010) used an electrical circuit analogy to model the fracture related pressure drop. Carey et al. (2015) extended the model for multiple hydraulic fractures by lumping multiple fractures into an effective fracture. Ma et al. (2019) proposed a new model to simulate the pressure transient in wellbore by incorporating the

perforation friction and pressure-dependent fluid leak-off. A workflow was developed to interpret the water hammer data for fracture geometry characterization using an iterative ensemble smoother algorithm.

In this paper, a water hammer model that describes the pressure transient in hydraulically fractured horizontal well is established. In this model, a new multi-cluster hydraulic fracture boundary condition is derived by considering the perforation friction and leak-off in fractures. The method of characteristics is applied to efficiently solve the water hammer model. The water hammer signals can be effectively simulated. Meanwhile, an interpretation workflow is developed based on the Markov Chain Monte Carlo (MCMC) method, which can generate the posterior statistics of fracture parameters (e.g., width, length, height). Then a series of synthetic and field cases are used to testify the performance of the proposed approach in hydraulic fracture characterization using water hammer data.

2. MODEL AND METHODOLOGY

2.1. Water Hammer Model

The water hammer model mainly consists of the wellbore pressure transient model, initial condition, and boundary condition. The pressure transient in wellbore is governed by the equations of continuity and motion:

$$\frac{\partial Q}{\partial t} + gA \frac{\partial H}{\partial x} + \frac{fQ|Q|}{2DA} = 0 \quad (1)$$

$$\frac{\partial H}{\partial t} - \frac{Q}{A} \sin \theta + \frac{a^2}{gA} \frac{\partial Q}{\partial x} = 0 \quad (2)$$

where A is the cross section area; Q is the flow rate; D is pipe diameter; H is the pressure head; f is the friction factor of pipe; a is the wave velocity, which can be calculated as

$$a = \sqrt{\frac{K/\rho}{1 + \left[\left(\frac{K}{E} \right) \left(\frac{D}{e} \right) \right] c}} \quad (3)$$

where K is the bulk modulus of fluid; ρ is fluid density; E is the Young's modulus of pipe; e is the wall thickness of pipe; c is an empirical parameter.

Initial condition indicates the distributions of flow rate and pressure head along the wellbore at the beginning of pump shutdown. Steady-state flow is assumed, and the flow rate along wellbore is identical. The pressure head along wellbore can be calculated by the Darcy-Weisbach equation (Brown, 2003).

$$Q_i^0 = Q_0 \quad (4)$$

$$H_1^0 = H_0, \Delta H = f \frac{\Delta L Q^2}{2gDA^2} \quad (5)$$

where subscripts 1 and i denote the first and i -th wellbore segment; ΔL is the length of wellbore segment; the friction factor f can be calculated as (Chen 1979)

$$f = f' \sigma \quad (6)$$

$$\frac{1}{\sqrt{f'}} = -2 \log \left[\frac{\varepsilon}{3.7065D} - \frac{5.0452}{\text{Re}} \right] \cdot \log \left(\frac{1}{2.8257} \left(\frac{\varepsilon}{D} \right)^{1.1098} + \frac{5.8506}{\text{Re}^{0.8981}} \right) \quad (7)$$

where σ is the ratio of resistance reduction; f' is the friction factor of water. ε is the pipe roughness; Re is the Reynold number.

The boundary conditions are derived for wellhead and hydraulic fracture, respectively. For wellhead, the step-wise decrease of flow rate is commonly applied, which can be expressed as

$$Q_1^t = Q(t) \quad (8)$$

For hydraulic fractures, it is assumed that multiple perforations may be open in one cluster, but only one hydraulic fracture can be generated, as shown in Fig. 1. Then the boundary condition of hydraulic fracture can be obtained by combining the correlations of perforation friction (Massaras et al. 2007) and fluid leak-off in hydraulic fractures.

$$P_w - P_f = \frac{8\rho Q^2}{\pi^2 N_p^2 d^4 C_D^2} \quad (9)$$

$$2N_f h_f l_f \frac{dw_f}{dt} = Q - Q_{leak} \quad (10)$$

where P_w is the bottomhole pressure; P_f is the hydraulic fracture pressure; N_p is the number of opened perforations; d is the diameter of perforation; C_D is the discharge coefficient; N_f is the number of fractures; h_f is fracture height; l_f is fracture half-length; w_f is fracture aperture; Q_{leak} is fluid leak-off rate. The fracture aperture and fluid leak-off rate can be calculated as (Carter 1957; Paige et al. 1995)

$$w_f = \frac{2(P_f - \sigma_{\min})h_f(1-\nu)}{\pi G \alpha_s} \quad (11)$$

$$Q_{leak} = \frac{2N_f h_f l_f C_L}{\sqrt{t_f}} \quad (12)$$

where $\sigma_{\min}$ is the minimum horizontal in-situ stress; ν is the Poisson ratio; G is the shear modulus; α_s is the shape

factor of fracture; C_L is the leak-off coefficient; t_f is the leak-off time.

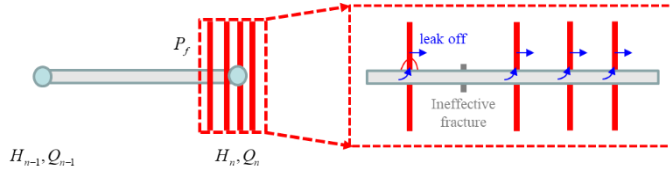


Fig. 1. Schematic of the hydraulic fractures in a stage.

2.2. Numerical Solution

The Method of Characteristics (MOC) is used to transform the partial differential equations, i.e., Eqs. (1) and (2), into ordinary differential equations along two characteristic lines:

$$C^+ : \begin{cases} \frac{dQ}{dt} + \frac{gA}{a} \frac{dH}{dt} + \frac{f}{2DA} Q|Q| - \frac{Qg}{aA} \sin \theta = 0 \\ \frac{dx}{dt} = a \end{cases} \quad (13)$$

$$C^- : \begin{cases} \frac{dQ}{dt} - \frac{gA}{a} \frac{dH}{dt} + \frac{f}{2DA} Q|Q| + \frac{Qg}{aA} \sin \theta = 0 \\ \frac{dx}{dt} = -a \end{cases} \quad (14)$$

The ordinary differential equations are solved by finite difference method after dividing the wellbore into n nodes. Then the pressure head H and flow rate Q at node i can be calculated as

$$H_i^{t+\Delta t} = \frac{C_P B_M + C_M B_P}{B_P + B_M} \quad (15)$$

$$Q_i^{t+\Delta t} = \frac{C_P - C_M}{B_P + B_M} \quad (16)$$

where $C_P = H_{i-1}^t + BQ_{i-1}^t$, $B_P = B + R|Q_{i-1}^t|$,

$C_M = H_{i+1}^t - BQ_{i+1}^t$, $B_M = B + R|Q_{i+1}^t|$, $B = \frac{a}{gA}$,

$$R = \frac{f \Delta x}{2gDA^2}, \Delta t = \frac{\Delta x}{a}.$$

Note that the pressure head and flow rate at the node 1 and n are calculated based on the boundary conditions of wellhead and hydraulic fracture.

2.3. Water Hammer Interpretation Method

In order to interpret the hydraulic fracture geometry, a MCMC-based inversion method is used to automatically history match the water hammer data. Firstly, the posterior probability density distribution function of the interpretation parameters is expressed as

$$P(m|o) \propto P(o|m)P(m) \quad (17)$$

where m denotes the interpretation parameters, including hydraulic fracture half-length, height, number of opened perforations, and number of fractures; o denotes the observed water hammer data. $P(o|m)$ and $P(m)$ are likelihood function and prior probability distribution function, respectively, which can be expressed as

$$P(o|m) = \frac{1}{(2\pi\sigma_n^2)^{\frac{M}{2}}} \exp \left\{ -\sum \frac{[o - f(m)]^2}{2\sigma_n^2} \right\} \quad (18)$$

$$P(m) = \frac{1}{(2\pi\sigma_m^2)^{\frac{M}{2}}} \exp \left[-\sum \frac{(m - \bar{m})^2}{2\sigma_m^2} \right] \quad (19)$$

where M is the number of observed data; σ_n and σ_m are the standard deviations of observed data and interpretation parameters.

The posterior probability density distribution function, i.e., Eq. (17), is used as the objective function to interpret the fracture parameters. Then MCMC method is applied to solve the objective function through efficient random sampling and acceptance/rejection operations. The optimal parameters of fracture geometry can be obtained, and the uncertainty about the interpreted parameters can be also evaluated.

3. RESULTS AND DISCUSSIONS

In this section, the interpretation method is firstly verified through a synthetic case. Then a multi-stage multi-cluster hydraulically fractured well in a shale gas reservoir is used to testify the applicability of the interpretation method in practical engineering problems.

3.1. Synthetic Case Study

A synthetic model is constructed with the basic parameters of wellbore, fluid, and reservoir, as shown in Table 1. We assume that there are five hydraulic fractures generated, and 24 perforations are opened in a stage. The half-length of the hydraulic fracture is 180 m, and the height is 30 m. Then the water hammer data is simulated through the proposed water hammer model, as shown in Fig. 2. The fracture half-length, height, number of hydraulic fractures, and number of opened perforations are used as the interpretation parameters.

Table 1. Basic parameters for the synthetic model

Parameter	Value
Vertical well length (m)	3180
Horizontal well length (m)	2120
Pipe diameter (mm)	118.6
Pipe wall thickness (mm)	10.54
Pipe Young's modulus (GPa)	206
Pipe Poisson ratio	0.3
Pipe wall roughness (mm)	0.015

Wellbore friction factor	0.05
Fluid density (kg/m ³)	1000
Fluid viscosity (mPa·s)	2
Fluid leak-off coefficient (m/s ^{0.5})	2×10^{-6}
Rock Young's modulus (GPa)	42
Rock Poisson ratio	0.22
Minimum in-situ stress (MPa)	76.1
Perforation diameter (mm)	8.128

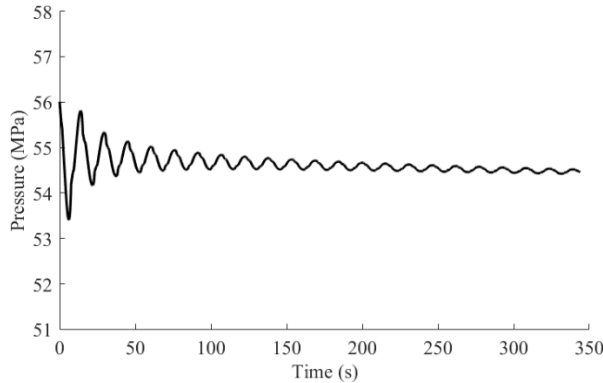


Fig. 2. Water hammer curve of the synthetic model.

The initial guess, as well as the lower and upper bounds, for the interpretation parameters are given as the input data, as shown in Table 2. Then the MCMC method is used to history match the water hammer data and interpret the fracture geometry. Fig. 3 shows the history matching results of water hammer data, and the interpreted parameters are listed in Table 3. The good agreement between the water hammer curves and the small difference between the interpreted and true parameters verify the effectiveness of the interpretation method. Moreover, the uncertainty evaluation of the interpretation parameter is given in Fig. 4, which shows the probable distribution of the fracture geometry parameters.

Table 2. Input data for the interpretation

Parameter	Initial guess	Lower bound	Upper bound
Half length (m)	150	50	200
Height (m)	25	20	50
Number of fractures	4	0	5
Number of opened perforations	20	0	40

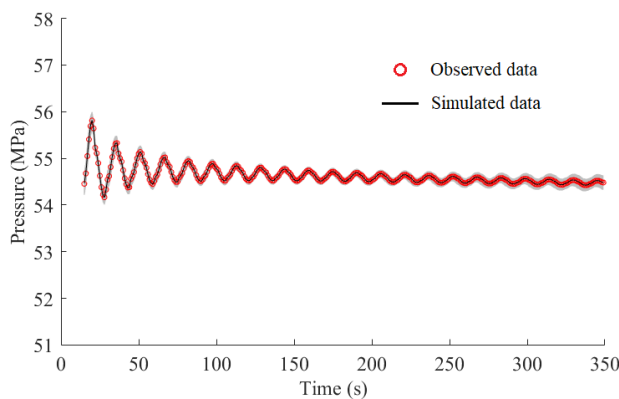
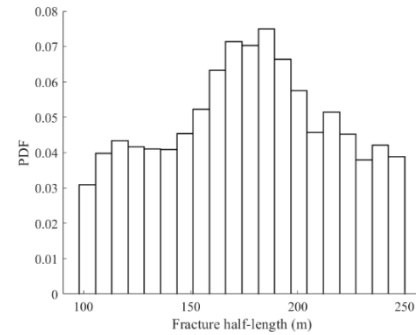


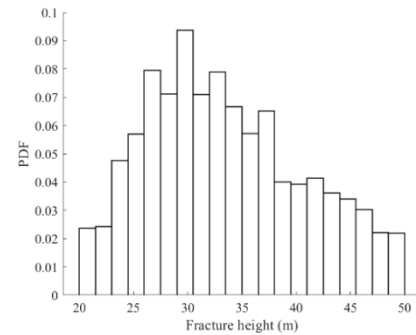
Fig. 3. History matching results of water hammer data.

Table 3. Comparison between the interpreted and true parameters

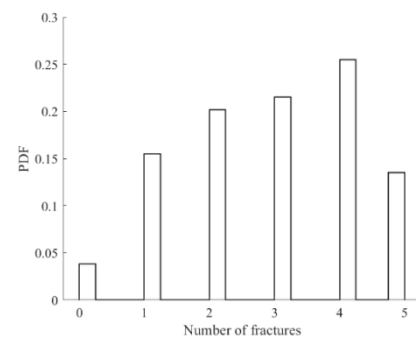
Parameter	Interpreted data	Standard deviation	True data
Half length (m)	164.43	44.52	180
Height (m)	32.83	6.80	30
Number of fractures	4	0.7	5
Number of opened perforations	23	1.6	24



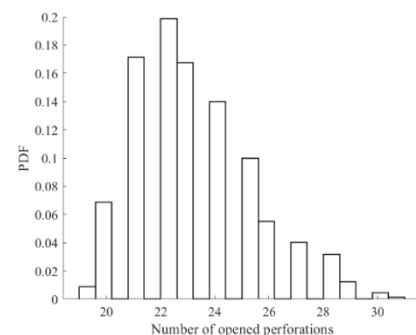
(a) Fracture half-length



(b) Fracture height



(c) Number of fractures



(d) Number of opened perforations

Fig. 4. Uncertainty evaluation of the interpretation parameter.

During hydraulic fracturing stimulation, other monitoring techniques can provide extra information about the fracture geometry. For example, the downhole imaging technology and fiber-optic sensing can accurately evaluate the opening of perforations. To evaluate the effect of the prior information on the interpretation results, we treat the number of fractures and number of opened perforations as known parameters. The input data for the fracture half-length and fracture height in Table 2 is used. Fig. 5 shows the history matching results of water hammer data with the constraints of the prior information, and Table 4 lists the interpreted parameters. Better fitting and interpretation results are obtained after the introduction of the prior information. This implies that the interpretation accuracy can be improved by using multiple monitoring techniques in the hydraulic fracturing processes to obtain more prior information.

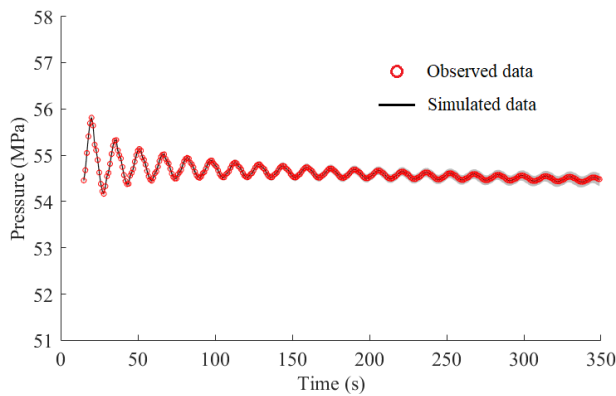


Fig. 5. History matching results of water hammer data with the constraints of the prior information.

Table 4. Comparison between the interpreted and true parameters with the constraints of the prior information

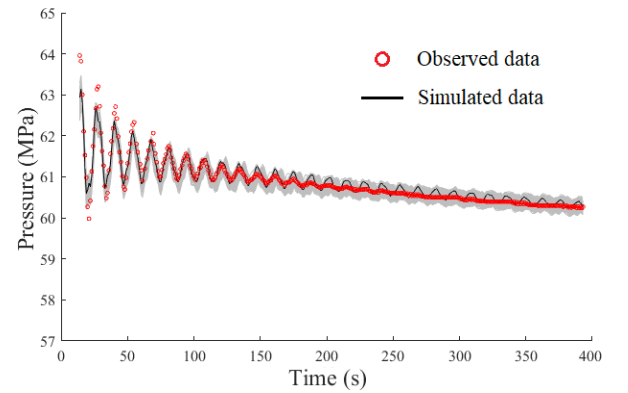
Parameter	Interpreted data	Standard deviation	True data
Half length (m)	179.55	43.05	180
Height (m)	30.19	4.47	30

3.2. Real Field Case Study

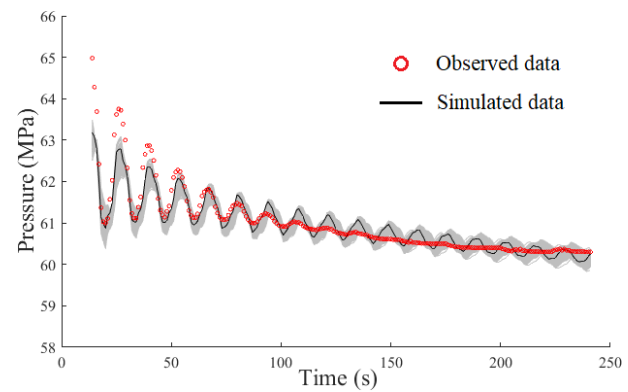
In this section, a real field case of a multi-stage hydraulic-fractured horizontal well in a shale gas reservoir is used to demonstrate the applicability of the proposed method. The length of the vertical section of the wellbore is 33445 m, and the length of the horizontal section is 1818 m. The horizontal is divided into 20 fracturing stages with the length of each stage being 90.9 m. 9 clusters for each stage and 7 perforations for each cluster are designed. One or two temporary plugging are applied for each stage to improve the fracturing efficiency. Then based on the water hammer data of each stage, the parameters of hydraulic fracture geometry are interpreted through the proposed approach.

We take the fifth and sixth stages as examples to show the history matching results of water hammer data, as shown

in Fig. 6. Good agreements between simulated data and observed data of real field cases can be still achieved, which testify the performance of the proposed method in hydraulic fracture characterization. Besides, the interpretation results of fracture geometry for all stages are summarized in Fig. 7. The averaged fracture half-length and height are 131.87 m and 25.8 m, respectively. Moreover, the fracture length and height become larger after plugging, which proves the effectiveness of plugging.

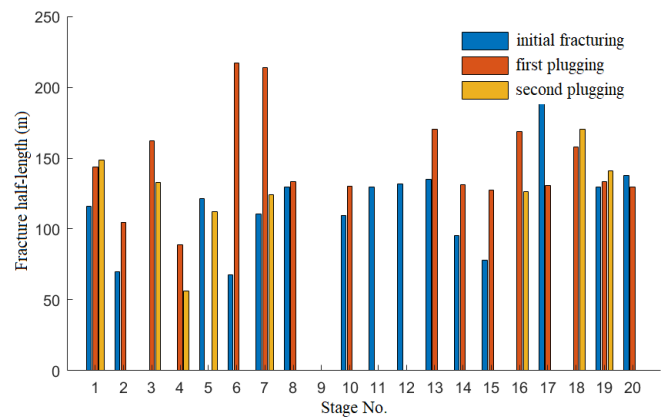


(a) Fifth stage

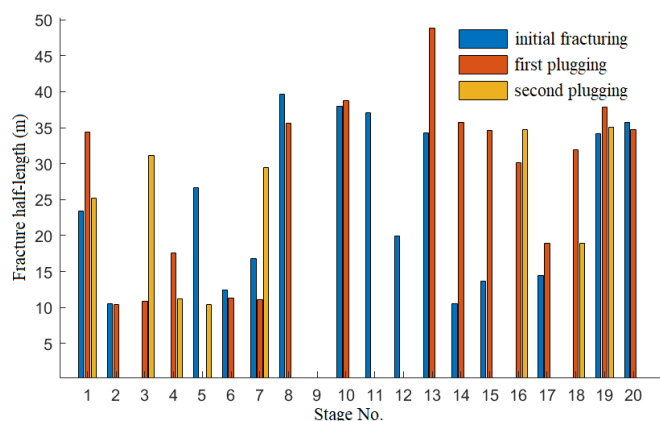


(b) Sixth stage

Fig. 6. Results of water hammer data history matching of the fifth and sixth stages.



(a) Fracture half-length



(b) Fracture height

Fig. 7. Interpretation results of fracture geometry for all stages.

4. CONCLUSIONS

In this paper, a novel hydraulic fracture diagnostics approach using the water hammer data is presented. A water hammer model that describes the pressure transient in hydraulically fractured horizontal well is established by fully considering the effect of wellbore friction, perforation friction, and fluid leak-off in hydraulic fractures. A hydraulic fracture geometry interpretation approach is developed based on the MCMC method.

The proposed approach is verified by a series of synthetic models. The results show that the parameters of fracture half-length, height, number of effective fractures, and number of opened perforations can be effectively identified. Besides, the introduction of prior information about the fracture geometry can improve the interpretation accuracy.

Finally, a real field case of a multi-stage hydraulic-fractured horizontal well in a shale gas reservoir is used to testify the applicability and efficiency of the proposed approach.

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